

Athishka Thushan

Temporary Lecturer, Department of Mechanical Engineering, University of Moratuwa

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PROFILE AND RESEARCH INTEREST

Mechanical Engineering graduate specialized in Mechatronics, with strong research interests in **field robotics**, **robot manipulators**, **intelligent control** and **autonomous navigation**, particularly in applying decision-making techniques to mechatronic and robotic systems for addressing real-world engineering challenges. Possesses hands-on experience in mechanical and electronic system development through participation in Team FalconE Racing (Formula Student) and various academic and collaborative engineering projects.

EDUCATION

University of Moratuwa, Sri Lanka

Mar. 2021 – Jul. 2025

– **Bachelor of the Science of Engineering (Hons.)**

Specialized in Mechatronic Systems Engineering | Overall GPA: 3.75/4.00

Ananda College, Colombo 10, Sri Lanka

Jan. 2011 – Aug. 2019

– **GCE Advanced Level (A/L)**

Physical Science Stream (Combined Mathematics, Physics, Chemistry) | 3A Passes | Z Score: 2.46

– **GCE Ordinary Level (O/L)**

9A Passes

ACADEMIC EXPERIENCE

Temporary Lecturer

Aug. 2025 - Present

– *Department of Mechanical Engineering, University of Moratuwa, Sri Lanka*

Engaging for the lectures, practicals, tutorials, and workshops for undergraduates in Control Systems, Embedded Systems, Mechatronic System Design, Instrumentation and Automation Systems, MEMS/NEMS and Automotive Engineering.

Research Intern

Feb 2024 - May 2024

– *Shandong University of Science and Technology, Qingdao, China*

Conducted hands-on studies in laser processing and 3D printing, collaborating across the Robotics, Mining, and Mechatronics laboratories. Contributed to faculty-supervised research projects on the study and application of magnetorheological fluids.

Projects



WS Ray - Underwater Drone for Immersive Reef Inspection

2025

Designed a bio-inspired ROV from first principles for delicate shallow-reef exploration, featuring a hull modeled on whale shark morphology and optimized via CFD. Incorporated fuzzy adaptive sliding-mode control for robust and precise real-time 6-DoF maneuverability.

Line and Wall Following Mobile Platform with Vision Based Sign Detection

2025

Built a Raspberry Pi-based robot with line following, wall following, and remote control modes, switchable via a mobile app. Integrated real-time road sign detection (Random Forest + OCR) across all modes for dynamic speed and navigation decisions with 0.99 accuracy, fully optimized for CPU-only operation.

Smart Mobile Robot for Warehouse Managing

2024

Designed a smart mobile robot with mecanum drive and a 4-DOF manipulator with suction gripper for autonomous pick-and-place. Implemented in ROS 2 with SLAM Toolbox for mapping, Navigation2 for path planning, sensor fusion and a web-based teleoperation interface with live video.

MEMS Based Microgripper for Biological Cell Manipulation

2024

Designed a MEMS microgripper integrating PZT actuation, a bridge-type amplifier, and capacitive sensing for precise, force-aware manipulation of 10–100 μm cells. Validated performance through COMSOL simulations of voltage–displacement response and nano–micro newton sensitivity.

Trimming Sorting Packing Automation Project

2023

Collaborated on an industrial automation project to automate the post-processing line of a glove manufacturing plant, focusing on trimming, sorting, and packing stages to enhance efficiency and consistency.

IOT Based Environment Monitoring System2023

Developed a multi-sensor IoT station measuring temperature, humidity, light, solar irradiance, wind speed and direction (ultrasonic anemometer), and rainfall (tipping-bucket rain gauge). Integrated calibrated signal conditioning with real-time mobile and web dashboards for continuous environmental monitoring.

Smart Parking Lot2022

Built an automated parking system for constrained lots using computer vision-based license plate recognition and RFID access control integrated with a cloud-backed database for gate management. Implemented safety sensors and a mobile app for real-time occupancy tracking and user access.

PUBLICATIONS

[1] H. D. Ekanayake, M. D. A. Thushan, K. D. S. H. M. Perera, Y. W. R. Amarasinghe, and W. A. D. M. Jayathilake. “CFD-Driven Hydrodynamic Optimisation of a Whale Shark-Inspired Underwater Drone for Shallow Reef Observation”. In: *Proceedings of the 11th World Congress on Mechanical, Chemical, and Material Engineering (MCM’25) – HTFF 2025* (Aug. 2025). URL: <https://doi.org/10.11159/htff25.243>.

[2] H. D. Ekanayake, M. D. A. Thushan, K. D. S. H. M. Perera, Y. W. R. Amarasinghe, and W. A. D. M. Jayathilake. “Development of Marine-Grade Acrylonitrile Butadiene Styrene (ABS) Based Hull of an Underwater Drone”. In: *Proceedings of the Moratuwa Engineering Research Conference (MERCon 2025)*. Presented at MERCon 2025, Best Paper Nominee – Mechanical Engineering Systems. Moratuwa, Sri Lanka: IEEE, Aug. 2025. URL: <https://doi.org/10.1109/MERCon67903.2025.11217039>.

[3] K. D. S. H. M. Perera, M. D. A. Thushan, H. D. Ekanayake, Y. W. R. Amarasinghe, and W. A. D. M. Jayathilake. “A Fuzzy-Adaptive Sliding Mode Control Strategy for Optimizing Control of Underwater Drone”. In: *Proceedings of the 2025 Moratuwa Engineering Research Conference (MERCon 2025)*. Presented at MERCon 2025. Moratuwa, Sri Lanka: IEEE, Aug. 2025. URL: <https://doi.org/10.1109/MERCon67903.2025.11217090>.

[4] M. D. A. Thushan, K. D. S. H. M. Perera, H. D. Ekanayake, Y. W. R. Amarasinghe, W. A. D. M. Jayathilake, H.A.G.C. Premachandra, and U-Xuan Tan. “WS Ray: Extending Human Perception with a Teleoperated Underwater Drone for Subaquatic Environments”. Manuscript accepted to 2026 IEEE 22nd International Conference on Automation Science and Engineering (CASE) - First Author and Presenter. 2026.

WORKING EXPERIENCE

Team FalconE Racing, University of Moratuwa, Sri LankaJul. 2023 – Jul. 2025

– Certified Electrical System Officer

Completed the EV Safety Level 3E certification and led the design of Sri Lanka’s first 600 V Li-Po battery pack for Formula Student UK 2026. Developed high-voltage architecture compliant with international EV safety standards, and conducted testing and maintenance for performance reliability.

– Electronics Engineer

Contributed to the implementation of the vehicle powertrain and telemetry systems, encompassing LV/HV wiring, PCB design, sensor integration/calibration, CAN communication and safety features.

Dipped Products PLC, Sri LankaNov. 2023 – Feb. 2024

– Project Engineer - Training

Engaged in automation projects, designing mechanical systems and accessories, including heat exchangers, duct systems, formers, and mountings. Gained hands-on exposure to industrial automation processes, enhancing technical expertise and understanding of core engineering standards and manufacturing practices.

SKILLS

Programming Languages:	Python, microPython, C, C++, LabVIEW, SQL, Latex
Robotics Platforms:	ROS2, MicroROS, Gazebo, WeBots
Embedded/Edge Platforms:	Raspberry Pi, ESP32, Atmel Studio, Siemens TIA Portal
Design, Modeling & Simulation:	SolidWorks, MATLAB, Simulink, Simscape, MSC Adams
FEA Tools:	ANSYS, COMSOL Multiphysics
Circuit Design & Simulation:	Altium Designer, Proteus, EAGLE
Languages:	English (Fluent), Sinhala (Native)

ACHIEVEMENTS

- **Team FalconE Racing, University of Moratuwa**
 - Engineering Design Event 8th Place – Formula Bharath 2024 Jan. 2025
 - Business Plan Presentation 4th Place – Formula Bharath 2024 Jan. 2025
 - Overall 10th Place EV category – Formula Bharath 2024 Jan. 2025
 - Best Looking Car – Formula Bharath 2024 Jan. 2025
- **Basketball, University of Moratuwa**
 - Champions in Sri Lanka University Games Aug. 2023
 - Champions in Inter University Games Sep. 2024
- **Basketball, Anada College, Colombo 10**
 - All Island Under 15 Basketball – B Division Champions Dec. 2014

VOLUNTEERING

University of Moratuwa

- Supervisor - Team Diyakawa for the Singapore Autonomous Underwater Vehicle Challenge (SAUVC) 2026
- Member - University of Moratuwa Mechanical Engineering Society (UMMES)
- Lab Facilitator - EXMO 2023 Exhibition University of Moratuwa
- Organizing committee member - MechatronX 2023

Ananda College, Colombo 10

- Science Society
- Old Anandian Engineers’ Guild Student Chapter (OAEG)

REFERENCES

Prof. Ranjith Amarasinghe ranama@uom.lk | (+94)71 8517882
Head of Department, Department of Mechanical Engineering, University of Moratuwa

Dr. Dumith Jayathilaka dumithj@uom.lk | (+94)11 2650301
Stream Coordinator, Mechatronic Systems Engineering Stream, Department of Mechanical Engineering, University of Moratuwa